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**Program:** Data Analytics with Generative AI — NIIT, Anna Nagar

# Capstone Project:

## Analyzing Food Delivery Demand Across Regions to Inform Market Expansion & Logistics

# Introduction

- As per the data, 100,000 food orders move through FoodTech's network across Urban, Semi-Urban, and Metropolitan India — each one a small decision made by a real customer, a delivery partner, and a restaurant.

- This project set out to answer a simple question with real evidence: as FoodTech grows, where should it grow next — and what should it fix first?

- What follows is not a collection of charts. It is a decision-ready story, built one hypothesis at a time — from raw data, through SQL and statistical testing, to a Power BI dashboard and predictive models — so that every recommendation in this deck is something the business can act on with confidence.

- **Five business objectives. Twenty tested hypotheses. One evidence-based roadmap.**

# Problem Statement:

FoodTech is a food-delivery platform operating across Urban, Semi-Urban, and Metropolitan markets — 100,000 orders, 22,000 customers, 1,100 delivery partners, and 550 restaurants. As the platform scales, leadership needs one evidence-based view of where demand is strongest, which customers and cuisines drive loyalty, how efficiently partners and restaurants operate, and how discounting and order failures affect profitability.

# Business Objective:

| # | Objective | Description |
|---|---|---|
| 1 | **Regional Demand & Market Expansion** | Identify which regions drive volume, revenue and repeat-order growth. |
| 2 | **Customer Behavior, Segmentation & Loyalty** | Understand how type, demographics and cuisine shape spend and retention. |
| 3 | **Delivery Partner & Logistics Efficiency** | Evaluate speed, ratings and delays across vehicle types and regions. |
| 4 | **Restaurant Performance & Operational Capacity** | Assess prep time, ratings and capacity utilization. |
| 5 | **Order Economics, Discounts & Profitability Risk** | Quantify how discounts, payment mode and cancellations affect revenue. |

# Data Description

**Domain:** Food-delivery logistics & customer analytics — Urban, Semi-Urban and Metropolitan markets

**Four connected source tables, joined into one unified analytical model:**

| Source Table | Rows | Columns | Role / Grain |
|---|---|---|---|
| **Orders** | 100,000 | 23 | Central fact table — one row per order |
| **Customers** | 22,000 | 10 | One row per customer |
| **Delivery Partners** | 1,100 | 10 | One row per delivery partner |
| **Restaurants** | 550 | 8 | One row per restaurant |

# Business Questions — Regional Demand & Customer Behavior

| OBJECTIVE 1 — REGIONAL DEMAND & MARKET EXPANSION | |
|---|---|
| Q1 | Which regions generate the highest total order volume and Net Revenue? |
| Q2 | How does average order value (Final_Amount) differ across regions? |
| Q3 | Which regions show the strongest repeat-order growth potential? |
| Q4 | Does order volume spike differently by region during festival periods? |

| OBJECTIVE 2 — CUSTOMER BEHAVIOR, SEGMENTATION & LOYALTY | |
|---|---|
| Q5 | How does Average Spend differ across Customer Type (New / Returning / Loyal)? |
| Q6 | Which cuisine is most associated with Loyal vs. New customers? |
| Q7 | Does Total Orders correlate with Average Spend at the customer level? |
| Q8 | How does Order Rating differ across customer segments? |
| Q9 | Which payment mode is preferred by each Customer Type? |
| Q10 | Does Age Group influence Preferred Cuisine choice? |

# Business Questions — Logistics, Restaurants & Economics

| OBJECTIVE 3 — DELIVERY PARTNER & LOGISTICS EFFICIENCY | |
|---|---|
| Q11 | How does Avg. Delivery Speed (KMPH) differ across Vehicle Type? |
| Q12 | Is there a relationship between Delivery Efficiency Score and Avg. Customer Rating? |
| Q13 | Do delayed deliveries differ significantly by region? |
| Q14 | Does Delivery Time (minutes) impact Order Rating? |

| OBJECTIVE 4 — RESTAURANT PERFORMANCE & OPERATIONAL CAPACITY | |
|---|---|
| Q15 | Does Avg. Preparation Time correlate with Avg. Rating at the restaurant level? |
| Q16 | Which cuisine type has the highest average restaurant rating? |
| Q17 | Do Low-Capacity-Flag restaurants show lower Avg. Rating than others? |
| Q18 | How does Order Capacity per Day vary by city? |

| OBJECTIVE 5 — ORDER ECONOMICS, DISCOUNTS & PROFITABILITY RISK | |
|---|---|
| Q19 | Does Discount Applied significantly affect Final Amount / Net Revenue per order? |
| Q20 | Is Order Status (Cancelled / Returned) associated with region? |

# Tools and Techniques Used

### SQL foundation (MySQL)

Joins, sub-queries, views, and stored procedures produced the first pass of grouped, aggregated answers directly from the relational data.

### Python EDA & statistical testing

Univariate, bivariate, and multivariate analysis (box plots, correlation heatmaps, cross-tabulations).

Followed by formal ANOVA, t-test, chi-squarent, chi-square, and Pearson correlation tests for all 20 hypotheses.

### Microsoft Excel & Validation

**PivotTables**, **PivotCharts**, and **advanced conditional formulas** (XLOOKUP, INDEX-MATCH) for rapid data cross-checking.

Built a supplementary spreadsheet dashboard to double-verify net revenue and promotional discount bands.

### Power BI dashboard

A 6-page interactive dashboard mirroring the same data model, built for business-facing, self-serve exploration.

### Machine learning

Feature-engineered regression (delivery time) and classification (order-failure risk) models.

Extending the analysis from 'what happened' to 'what's next'.

# Data Cleaning & Fixes Applied

### Region typo standardized

'Metropolitian' corrected to 'Metropolitan'.

City fields in Customers and Restaurants held the region tier, so they were renamed to Region for consistency.

### Column naming standardized

Standardized across all four tables so IDs join cleanly (e.g. resolving a BOM-corrupted Customer_ID column).

### Referential integrity validated

Validated end-to-end.

Row-count and orphan-record checks confirmed 0 orders with a missing customer, restaurant or delivery-partner link.

### Net_Revenue engineered

Engineered net of discounts.

Rating-based analysis scoped strictly to delivered orders for a like-for-like comparison.

### Skewed numeric fields

Log-transformed fields.

Outliers treated statistically ahead of hypothesis testing and modelling.

### Derived features added

Added Discount_Ratio, preparation-to-delivery time ratio, and weekend / festival flags.

# Possible Analysis

### Descriptive & Diagnostic Analysis

order volume, revenue, order value and delivery-time profiles by region, customer segment, vehicle type and restaurant

to establish what is happening and why.

### Statistical & Probability Analysis

ANOVA, independent t-tests, chi-square tests and Pearson correlation across all 20 business questions

each tested at $\alpha = 0.05$.

### Predictive Modeling

linear regression to predict delivery time

classification (Logistic Regression, Random Forest, XGBoost) to predict order-failure risk.

### KPI & Trend Analysis

headline platform KPIs plus monthly order trend and festival-flag seasonality

tracked through the Power BI dashboard.

# KPIs

| | | | |
|---|---|---|---|
| **100K** | **₹5.2 Cr** | **₹559.95** | **35.5 min** |
| Total Orders | Total Net Revenue | Avg. Order Value | Avg. Delivery Time |
| **4.02 / 5** | **89.8%** | **70%** | **93.03%** |
| Avg. Customer Rating | On-Time Delivery Rate | Repeat Customer Rate | Orders Delivered Successfully |

*KPIs derived from the Power BI Overview page across the full 100,000-order dataset, corresponding to the platform's headline scale, service quality and economics.*

SQL IMPLEMENTS AND ITS INSIGHTS

## Insights from SQL Scripts — Objective 1

Result Grid | Filter Rows: | Export:

| Region | Total_Orders | Total_Net_Revenue |
|---|---|---|
| Urban | 33421 | 17434107 |
| Semi-Urban | 33349 | 17355192 |
| Metropolitian | 33230 | 17273586 |

**Objective 1 · Total Orders & Net Revenue by Region**

→ Urban, Semi-Urban and Metropolitan are within ~1% of each other on both orders and revenue — demand is evenly split, not concentrated in one region.

## Insights from SQL Scripts — Objective 2

| Customer_Type | Avg_Spend | Customers |
|---|---|---|
| Loyal | 342.88 | 4400 |
| New | 326.66 | 6600 |
| Returning | 312.53 | 11000 |

**Objective 2 · Avg. Spend & Customers by Type**

→ Loyal customers spend the most per head (₹342.88) but Returning customers are the largest, most scalable base at 11,000 customers.

## Insights from SQL Scripts — Objective 3

| Pearson_Correlation_Efficiency_Rating |
|---|
| 0.3763572740098047 3 |

**Objective 3 · Efficiency Score ↔ Rating Correlation**

→ Pearson r = 0.376 (p<0.001) — a genuine, moderate link: delivery efficiency measurably moves the customer rating a partner earns.

# Insights from SQL Scripts — Objective 4

| | City | Avg_Capacity_Per_Day | Restaurants |
|---|---|---|---|
| | Metropolitan | 47.42 | 285 |
| | Urban | 19.15 | 217 |
| | Semi-Urban | 1.27 | 48 |

**Objective 4 · Avg. Capacity per Day by City**

→ Metropolitan restaurants run far higher daily capacity (47.4 orders/day) than Semi-Urban (1.3) — capacity, not demand, is the Semi-Urban ceiling.

# Insights from SQL Scripts — Objective 5

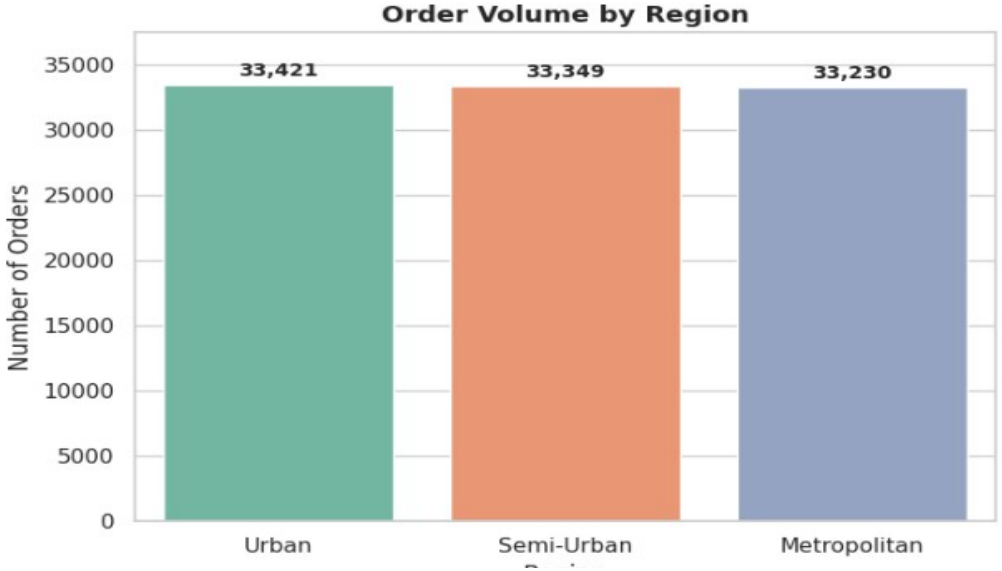
| Order_Status | Avg_Delivery_Time | Order_Count |
|---|---|---|
| Returned | 35.55 | 2022 |
| Delivered | 35.51 | 93027 |
| Cancelled | 35.45 | 4951 |

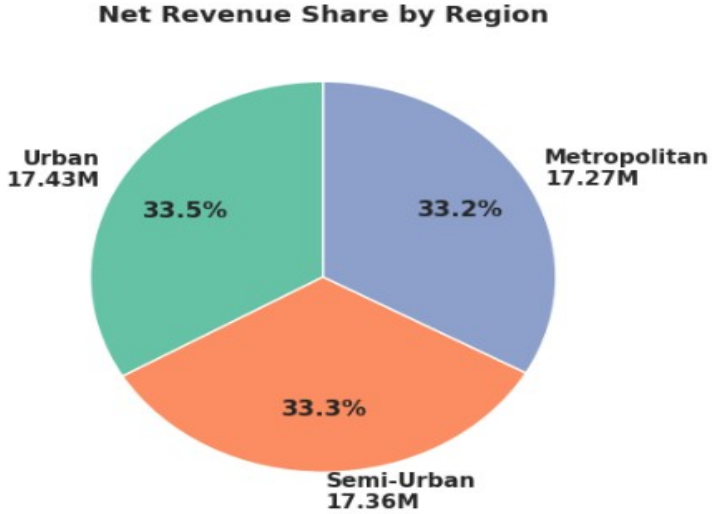
**Objective 5 · Delivery Time by Order Status**

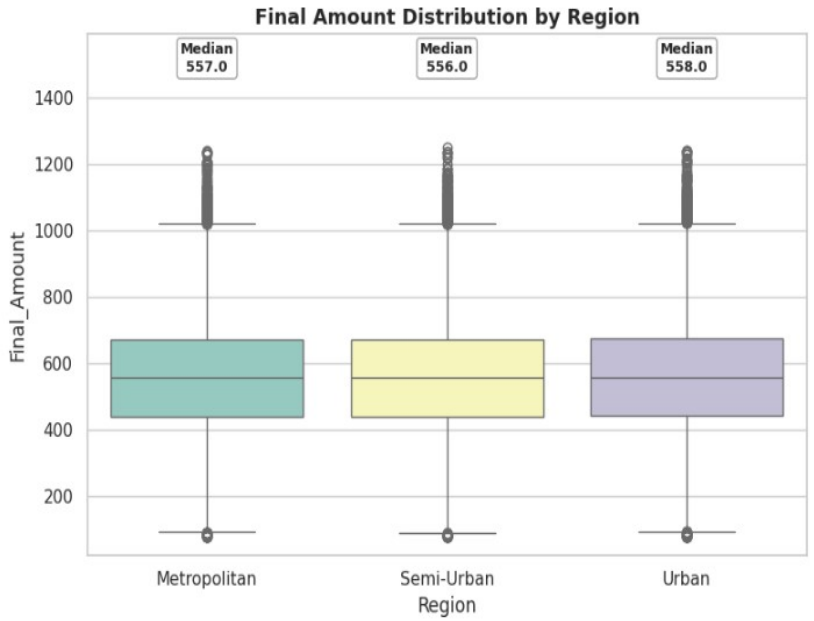
→ Delivered, Cancelled and Returned orders all average ~35.5 minutes — delivery time alone cannot flag a failing order, an early clue behind the ML classification result on Slide 22.

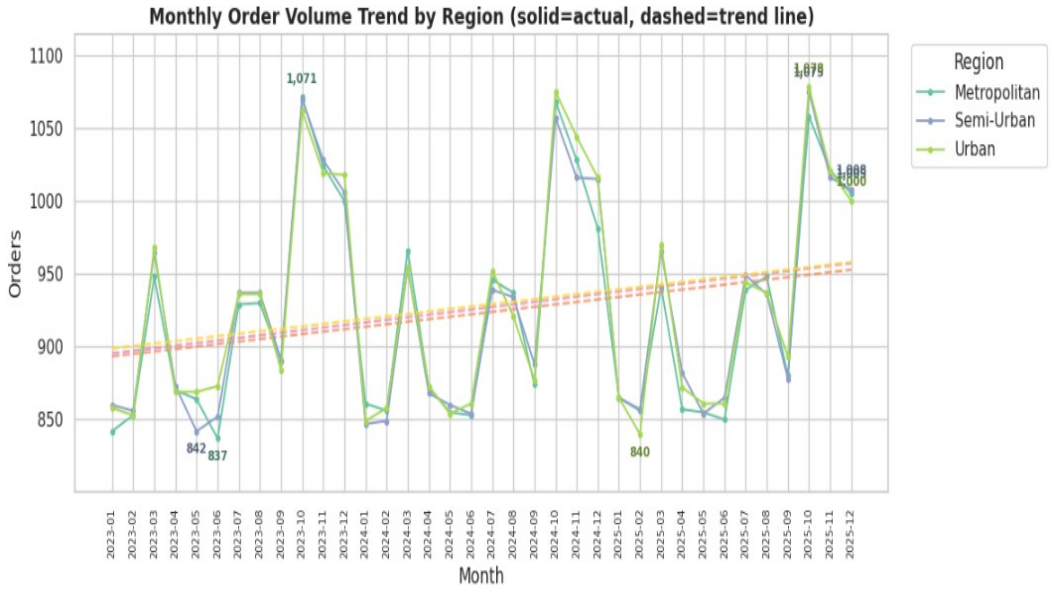
# Insights from EDA — Objective 1

**Objective 1 · Orders, Revenue Share & Monthly Trend by Region**



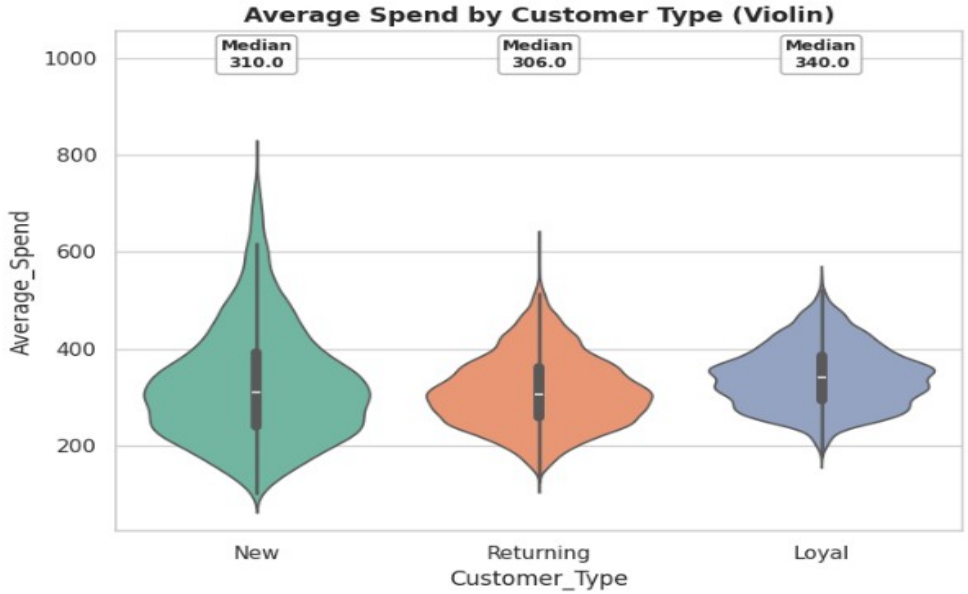



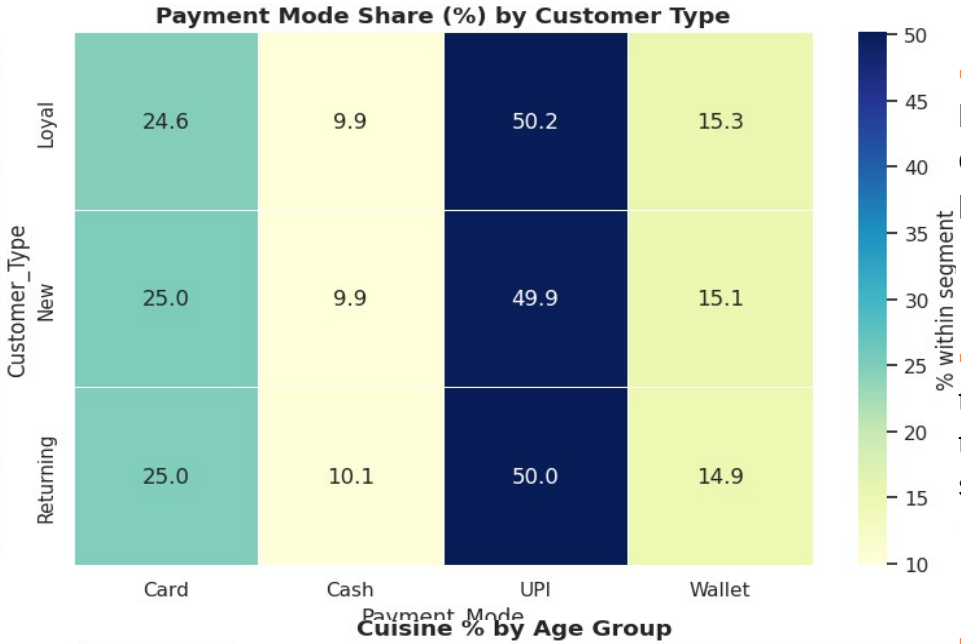




- Orders and revenue are split almost exactly one-third / one-third / one-third across Metropolitan, Semi-Urban and Urban — no single region dominates.
- Revenue share mirrors that split almost exactly (33.5% Urban, 33.3% Semi-Urban, 33.2% Metropolitan) — the pie chart confirms money follows orders region by region.
- Final_Amount distributions (box plot) are nearly identical across regions, confirming the ANOVA result on the Business Questions slide.
- Monthly order trend shows sharp, synchronized spikes across all three regions — a platform-wide seasonal/festival effect, not a regional one.

# Insights from EDA — Objective 2

## Objective 2 · Spend, Payment Mode & Cuisine by Segment

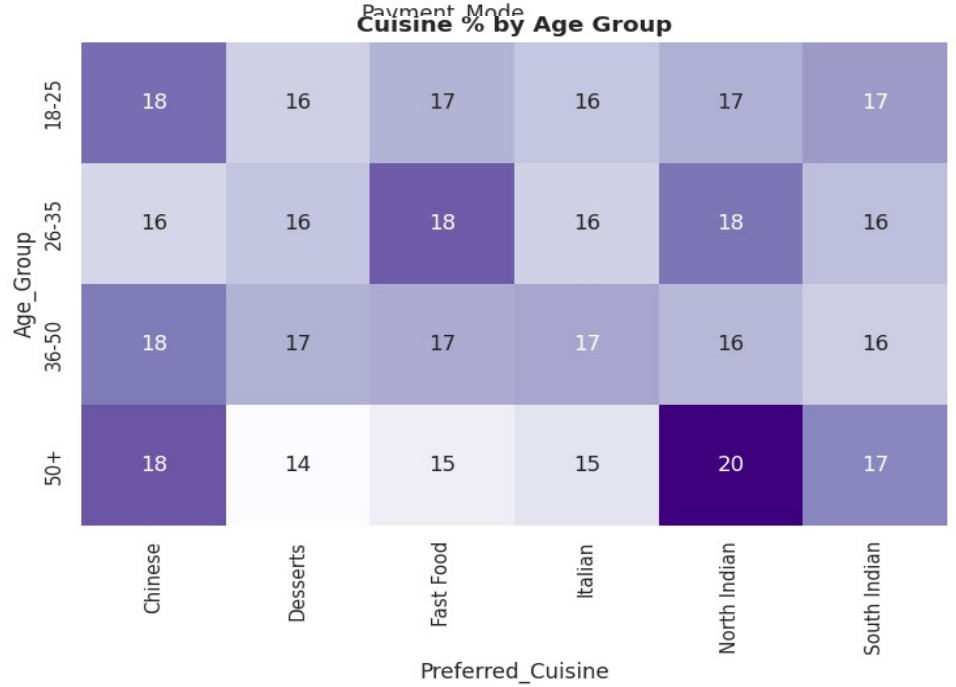



- Loyal customers show a visibly higher, tighter spend band than New or Returning customers in the box plot — loyalty compounds spend.

- UPI accounts for 50% of transactions across every customer type — payment preference is not a segmentation lever.



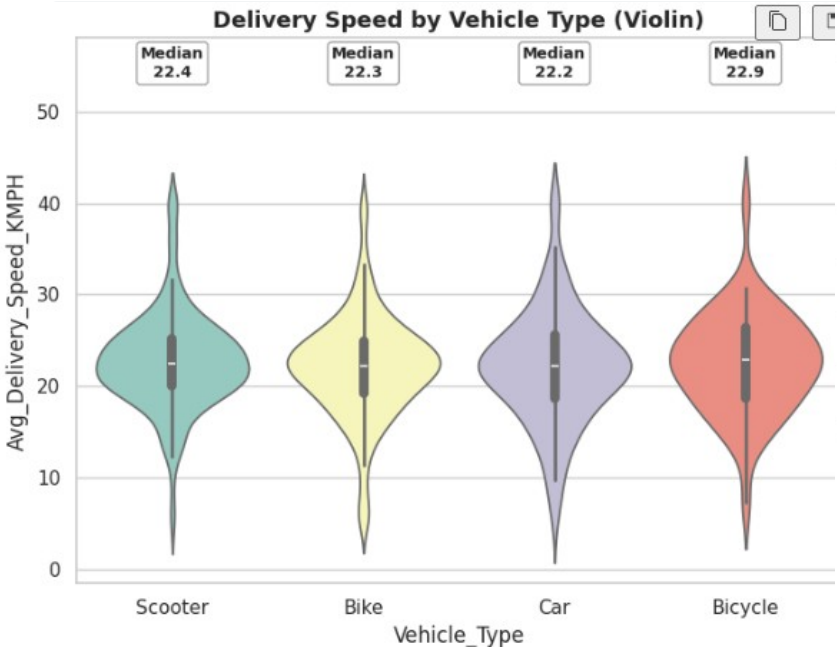



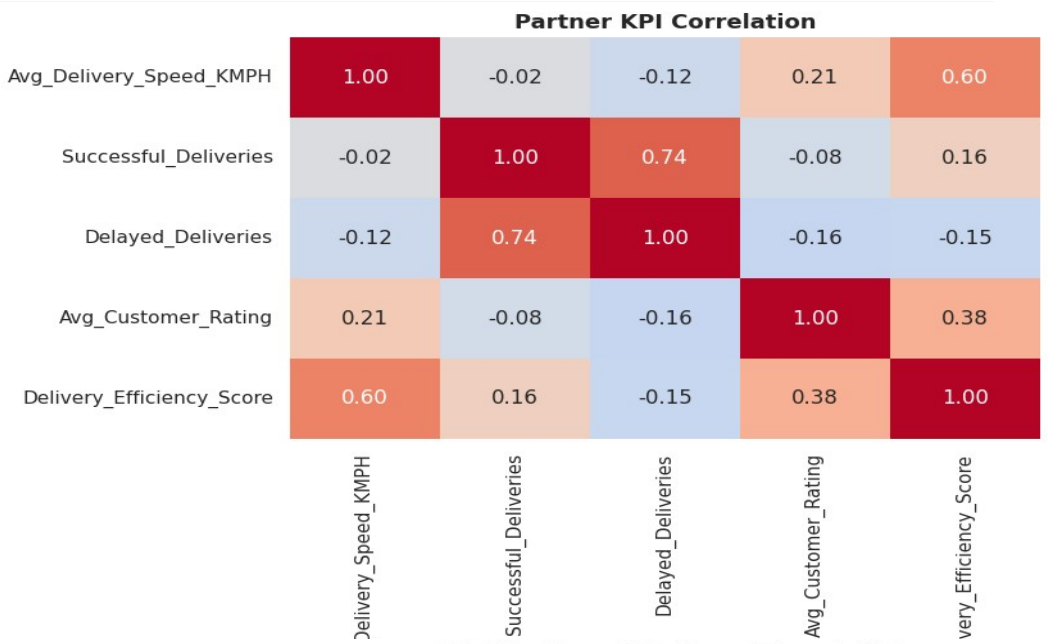

- Total Orders vs. Average Spend (scatter) shows only a faint upward trend — order frequency and basket size move almost independently of each other.

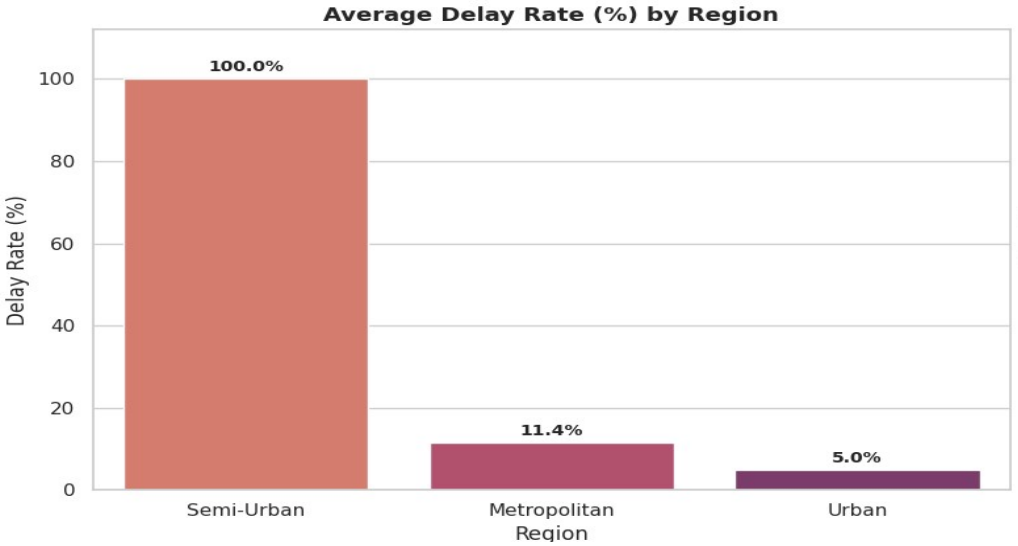
- Cuisine share by age group is fairly even (14–20% each), with a mild North Indian tilt among 50+ customers.

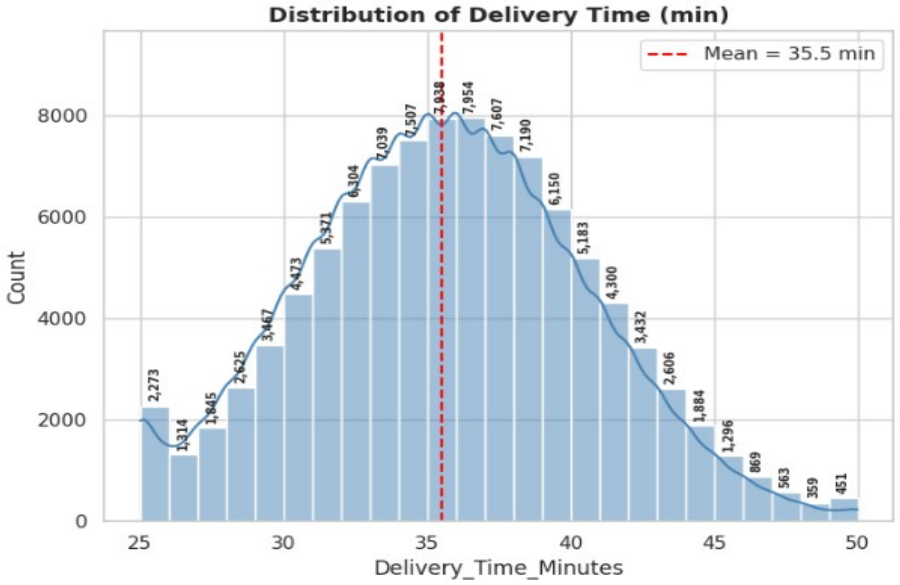
# Insights from EDA — Objective 3

## Objective 3 · Delivery Speed, Efficiency Correlation & Delay Rate



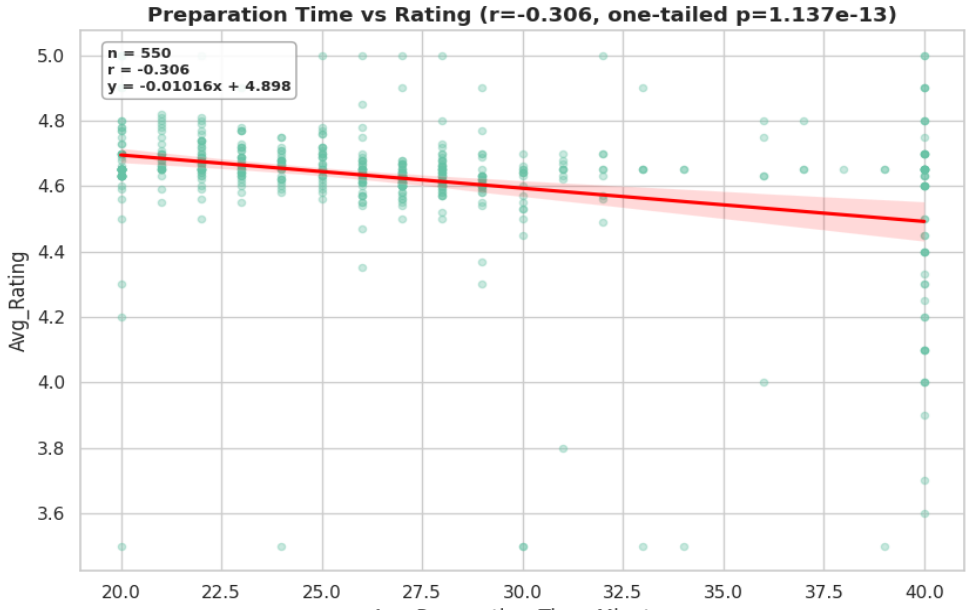



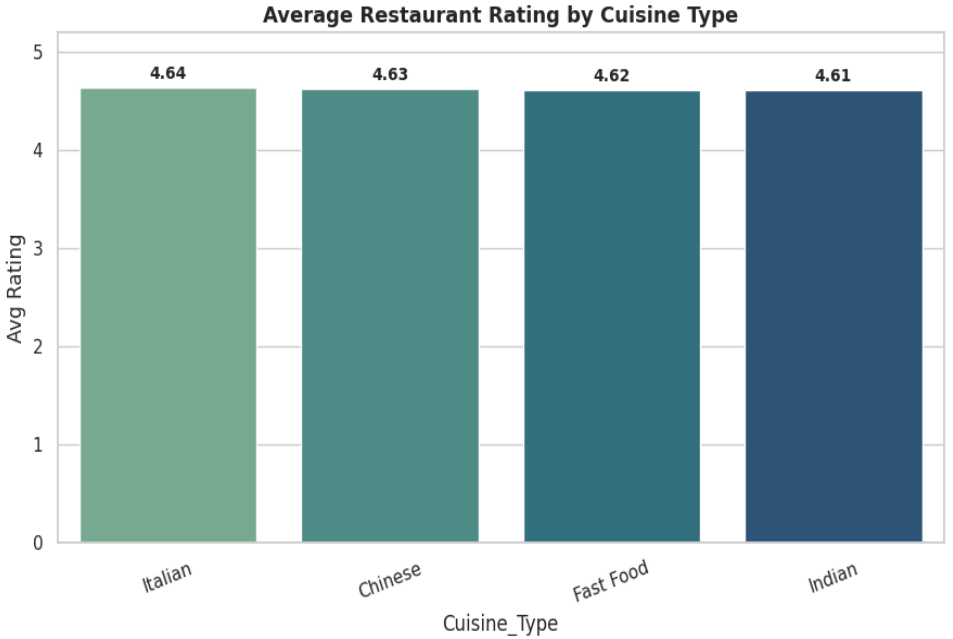



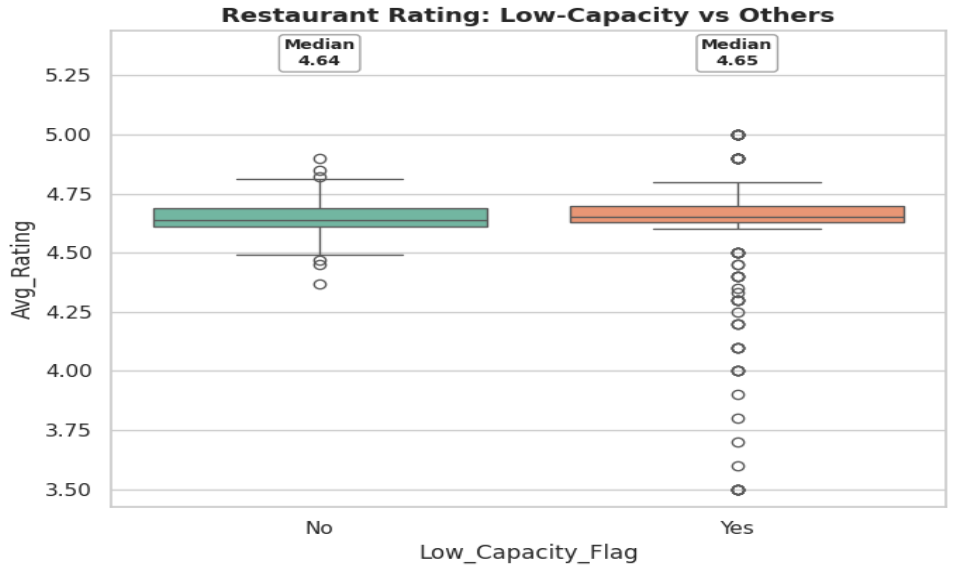

- Delivery speed is virtually identical across Scooter, Bike, Car and Bicycle — vehicle type is not the delay driver it might seem.
- The correlation heatmap confirms Delivery_Efficiency_Score is the strongest lever on Avg_Customer_Rating (r=0.60) and Avg_Delivery_Speed (r=0.60).
- Semi-Urban carries almost all of the platform's delay rate — a sharp, region-specific operational gap rather than a systemic one.
- Delivery time distribution is a smooth, single-peaked curve centered around 35 minutes — confirming the platform's average delivery time is a fair, representative figure, not skewed by outliers.

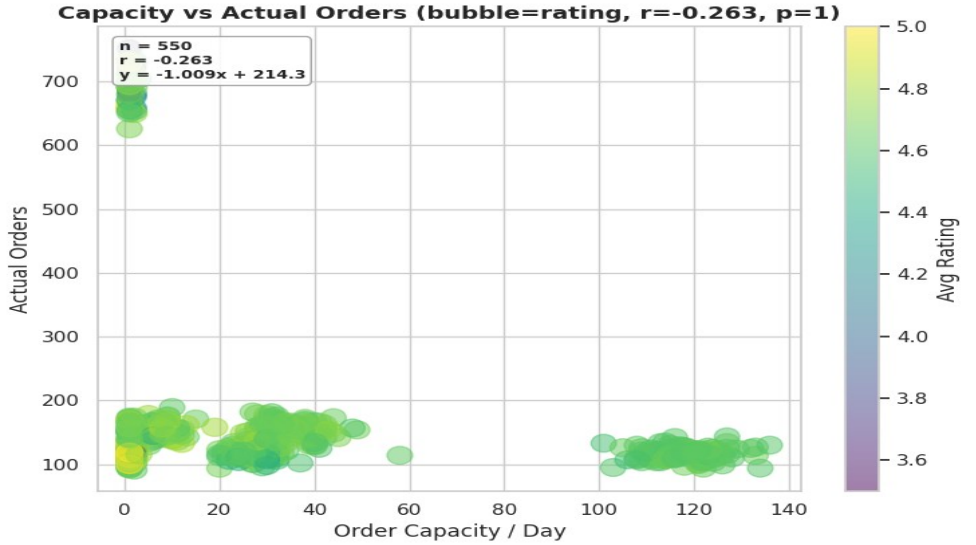
# Insights from EDA — Objective 4

## Objective 4 · Prep Time, Rating by Cuisine & Capacity



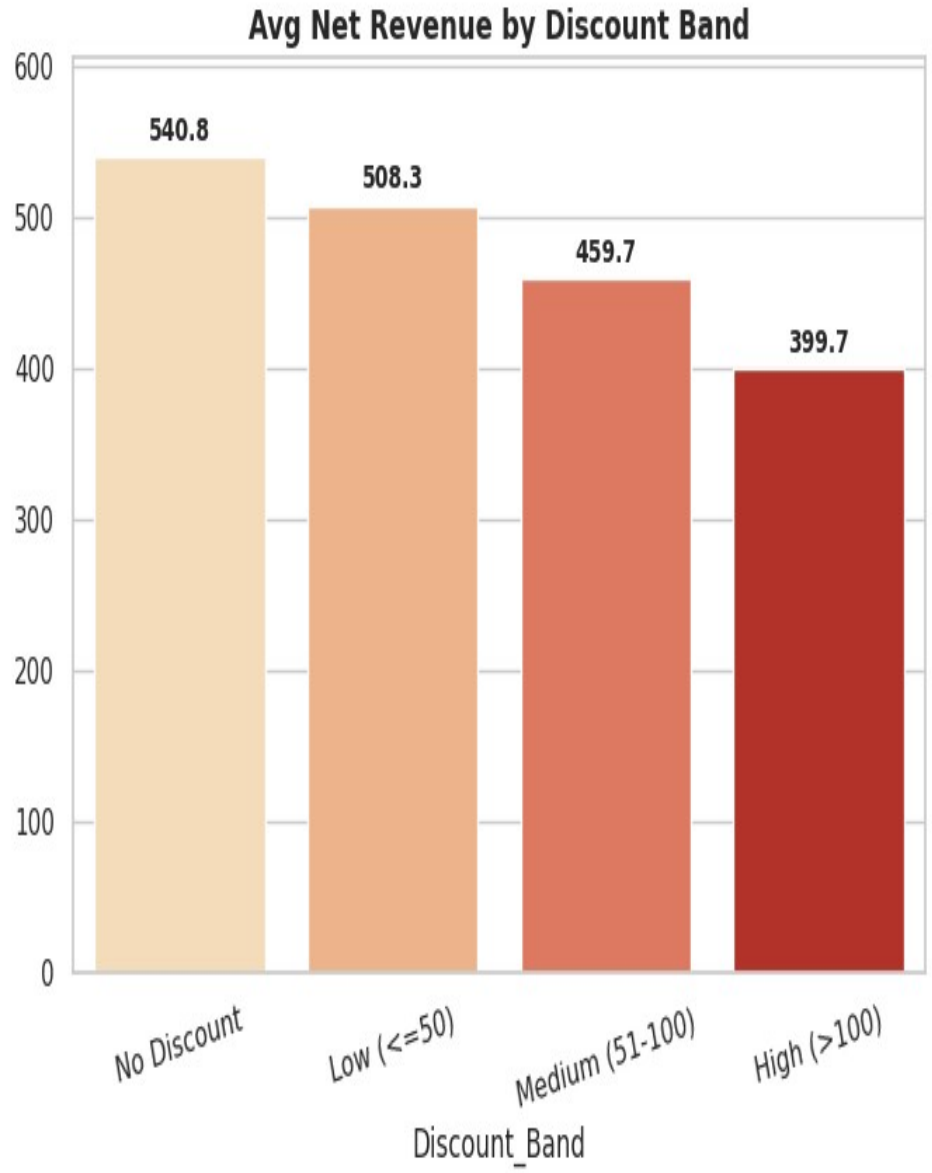



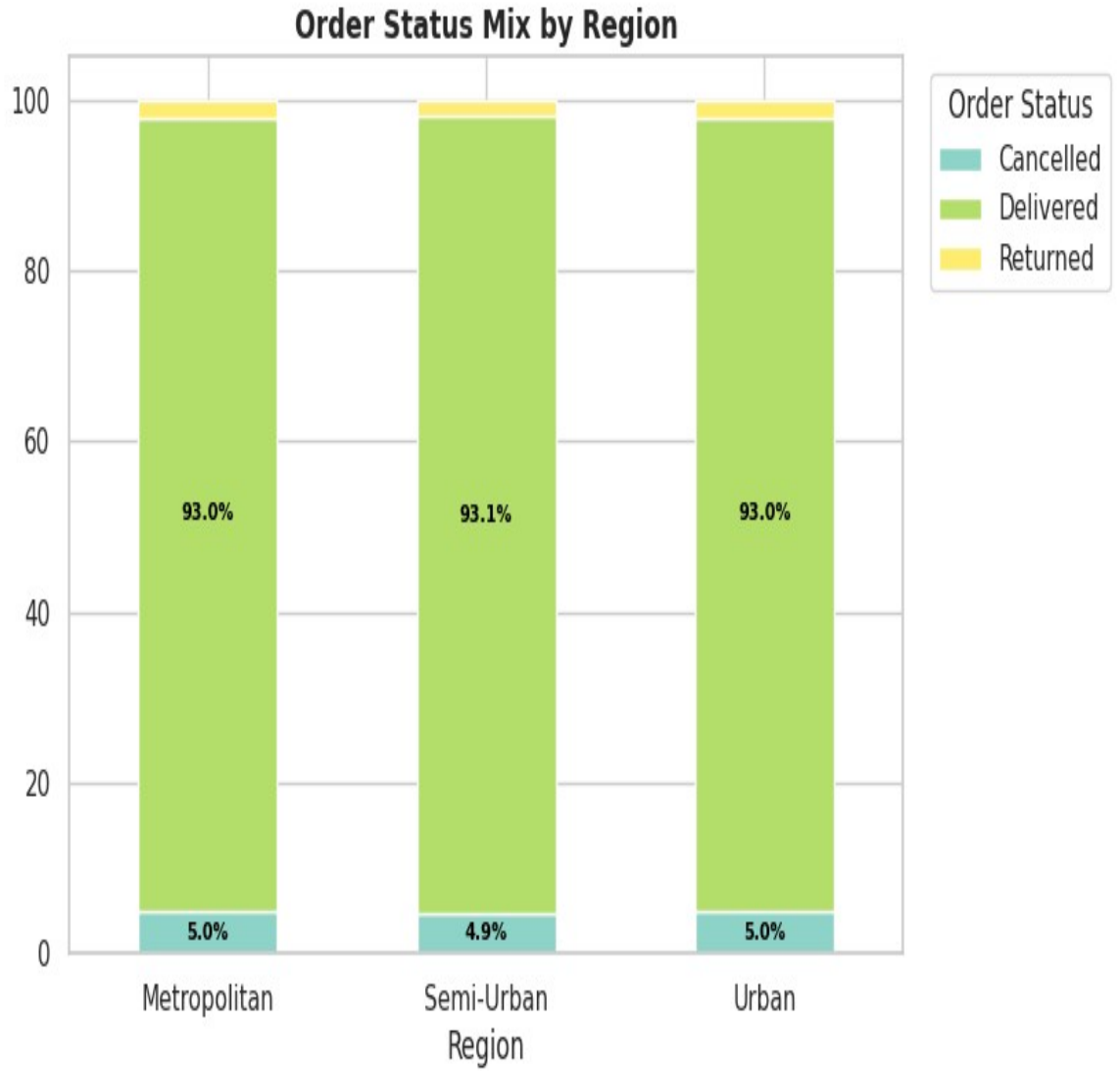




- Preparation time trends negatively against rating (r=−0.306) — restaurants that cook faster are rated modestly higher.
- Average rating is flat (~4.6) across all four cuisines — quality perception isn't a cuisine effect.
- The Low-Capacity-Flag box plot shows a real quality gap — flagged restaurants rate significantly lower on average (t=−3.03, p=0.001).
- Capacity vs. actual orders shows most restaurants clustered at low order counts regardless of stated capacity — supply, not demand, is the binding constraint.

# Insights from EDA — Objective 5

**Objective 5 · Discount Bands & Order Status by Region**



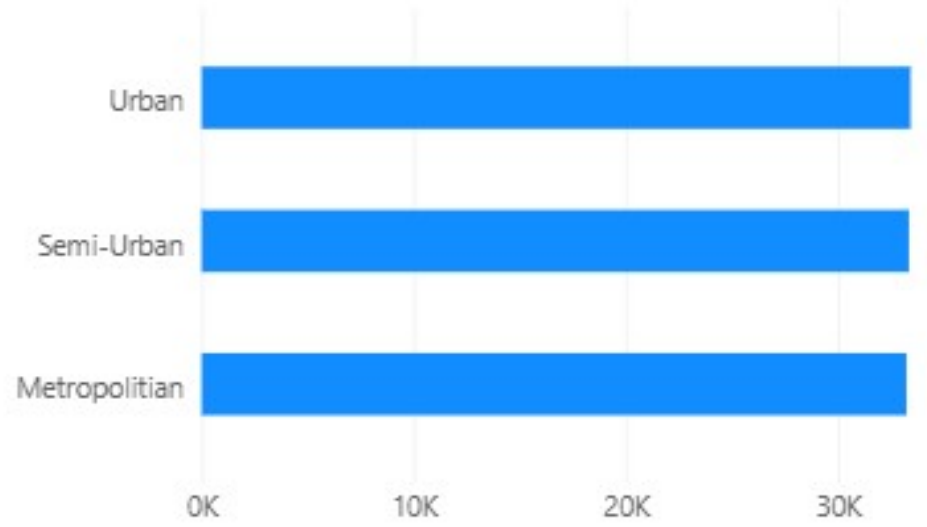

- Average net revenue per order falls in a clean, steady staircase from No Discount (₹540) to High discount (₹398) — every extra discount tier costs real revenue.
- Order-status mix (Delivered / Cancelled / Returned) is virtually identical across Metropolitan, Semi-Urban and Urban — cancellation risk is not regional.
- Cancellation rate holds steady at ~5% in every region — order-status risk isn't concentrated anywhere.

# POWER BI DASHBOARD 1

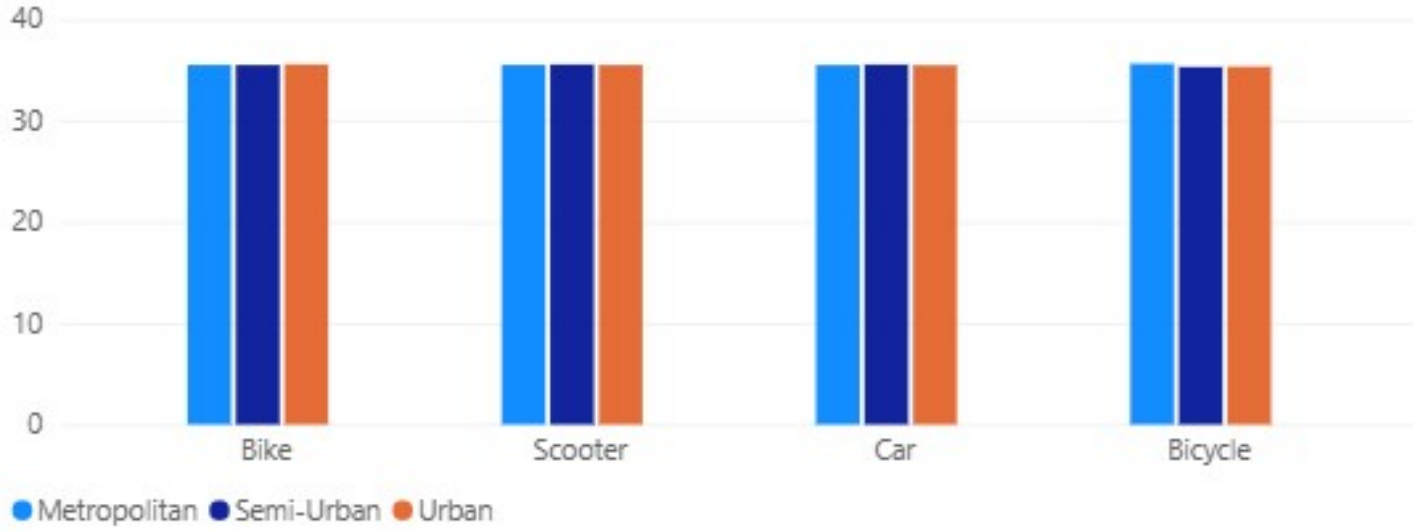
| Total Orders | Net Revenue | Average Order Value | Avg Delivery Time | Avg_Customer_Rating |
|---|---|---|---|---|
| 100K | 52M | 559.95 | 35.51 | 4.64 |

**Regional demand:** Order volume is evenly split across Urban, Semi-Urban, and Metropolitan regions (~33K orders each out of 100K total) — demand isn't concentrated in one region type, so market-expansion and resourcing decisions need to treat all three as equally important rather than prioritizing one.

## Total Orders
by Region




## Avg Delivery Time
by Vehicle_Type and City

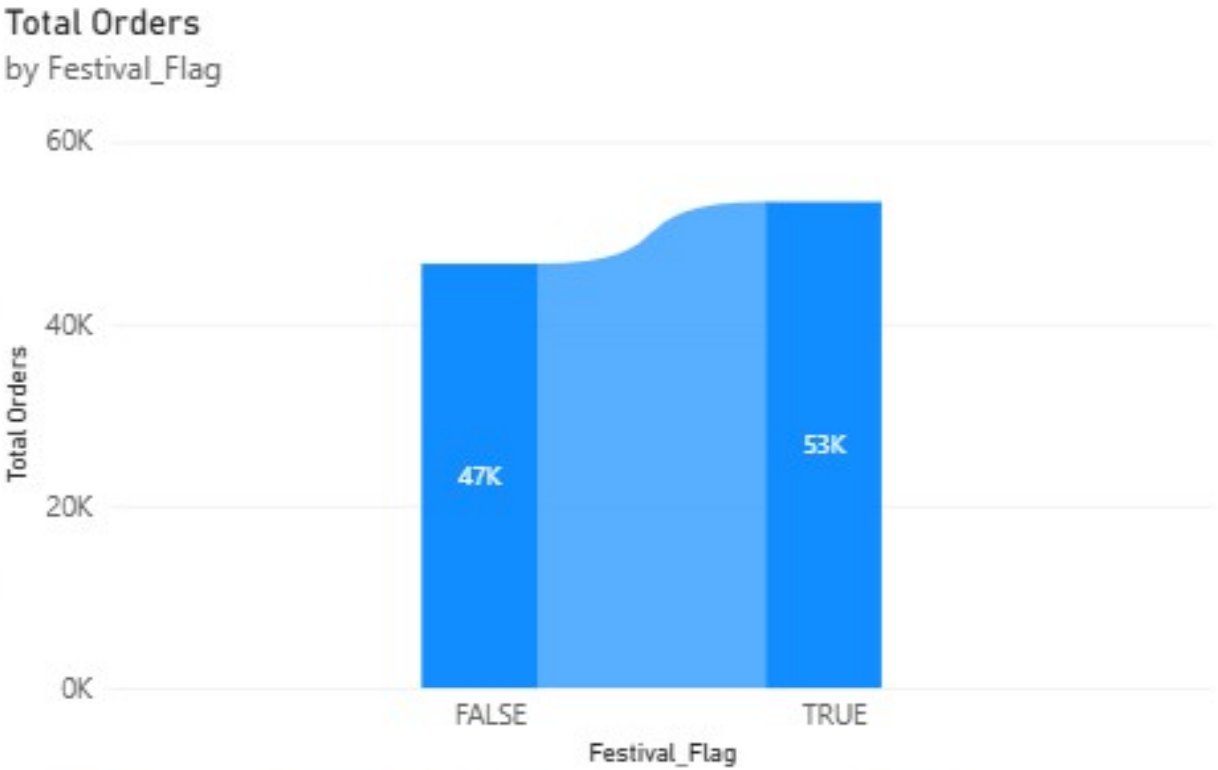



**Headline health check:** Against 100K orders and ₹52M net revenue, an average order value of ₹559.95, a 35.5-minute average delivery time, and a 4.02 average rating point to stable, consistent unit economics — no single number here is an outlier that demands urgent attention on its own.

**Delivery time by vehicle type:** Avg. delivery time holds steady at roughly 35 minutes regardless of vehicle type (bike, scooter, car, or bicycle) or city category. This means vehicle choice isn't a meaningful lever for speeding up deliveries — the ~35-minute average is likely driven by other factors (distance, order prep time, or traffic), not the delivery vehicle itself. Reallocating delivery partners toward "faster" vehicle types would not meaningfully cut delivery time based on this data

# POWER BI – DASHBOARD 2

**Seasonal pattern:** March, July, August, and October–December are flagged as festival months, and each shows meaningfully higher total orders (~9K–10K) compared to non-festival months (~7.5K–8K) — a clear ~20% seasonal demand uplift. This confirms festival periods are a predictable, exploitable driver of order volume, useful for staffing and delivery-partner scheduling decisions ahead of those months.







**Revenue distribution: Net Revenue** is nearly identical across Urban, Semi-Urban, and Metropolitan regions (~17M each) — consistent with the even order-volume split seen on the Overview page. There's no single region driving disproportionate revenue, meaning expansion or investment decisions shouldn't be skewed toward one region type over another based on this data alone.

# POWER BI – DASHBOARD 3

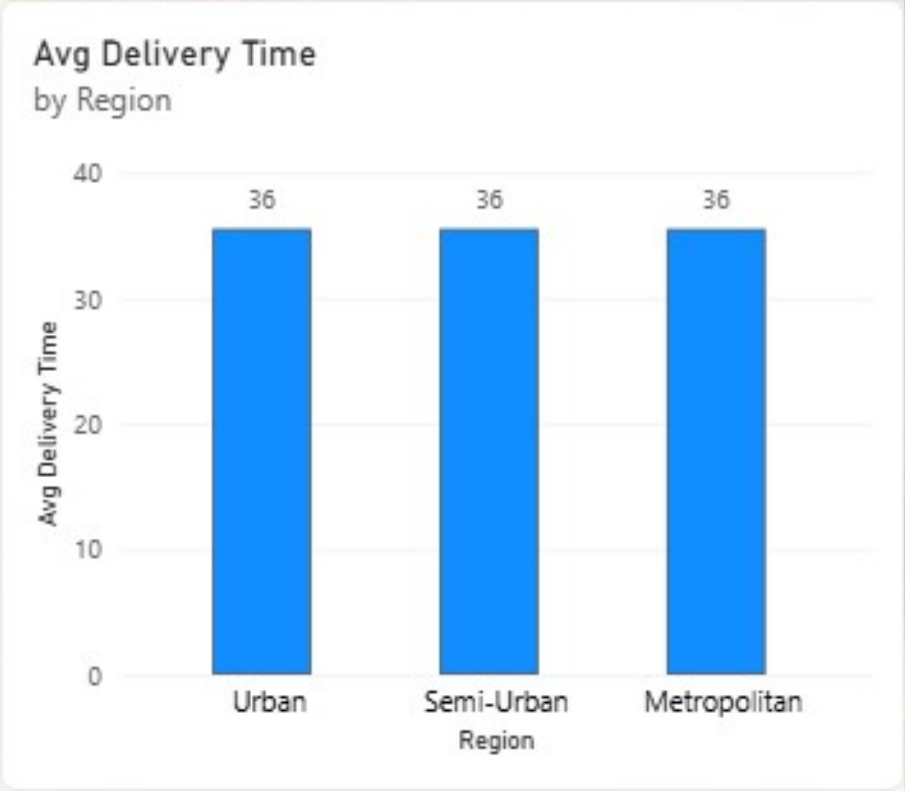
Avg Delivery Time
**35.51**

On-Time Delivery Rate
**89.82%**

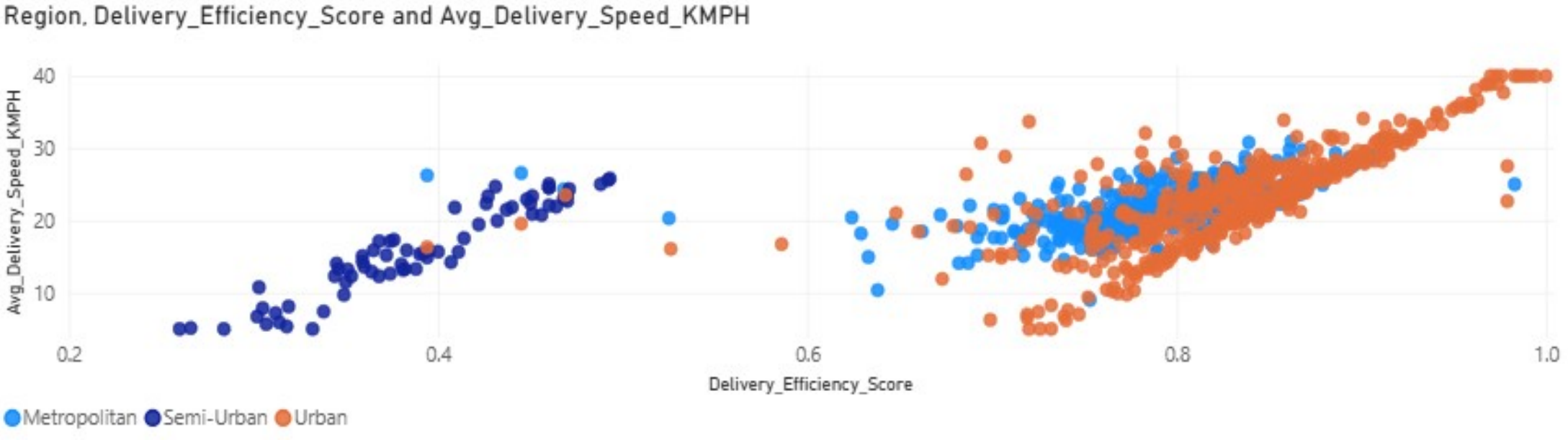
Cancellation Rate
**4.95%**

**Two-tier delivery partner performance:** The efficiency-vs-speed scatter reveals two distinct clusters rather than one smooth trend , a lower-performing group (efficiency ~0.2–0.5, speed under 20 KMPH), concentrated in Semi-Urban regions, and a stronger-performing group (efficiency 0.6–1.0, higher speed), dominated by Urban and Metropolitan partners.

**Regional consistency:** Avg. delivery time is virtually identical across Urban, Semi-Urban, and Metropolitan regions (~35–36 min each), reinforcing the Overview finding that region and vehicle type aren't meaningful levers for speeding up deliveries — the driver of delivery time lies elsewhere




**Service reliability:** A 89.82% on-time rate against a 4.95% cancellation rate indicates the service is generally reliable and the cancellation rate is the smaller of the two levers to investigate further

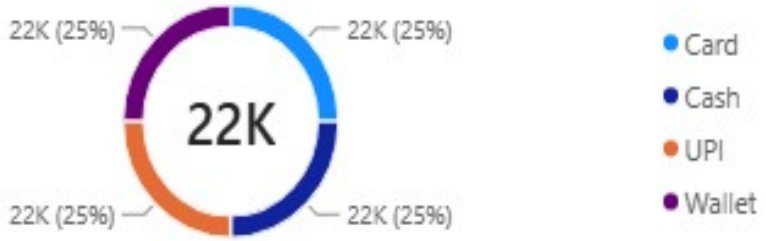


# POWER BI – DASHBOARD 4

| Average Order Value | Repeat Customer Rate | Average of Avg_Customer_Rating |
|---|---|---|
| 559.95 | 70.00% | 4.64 |

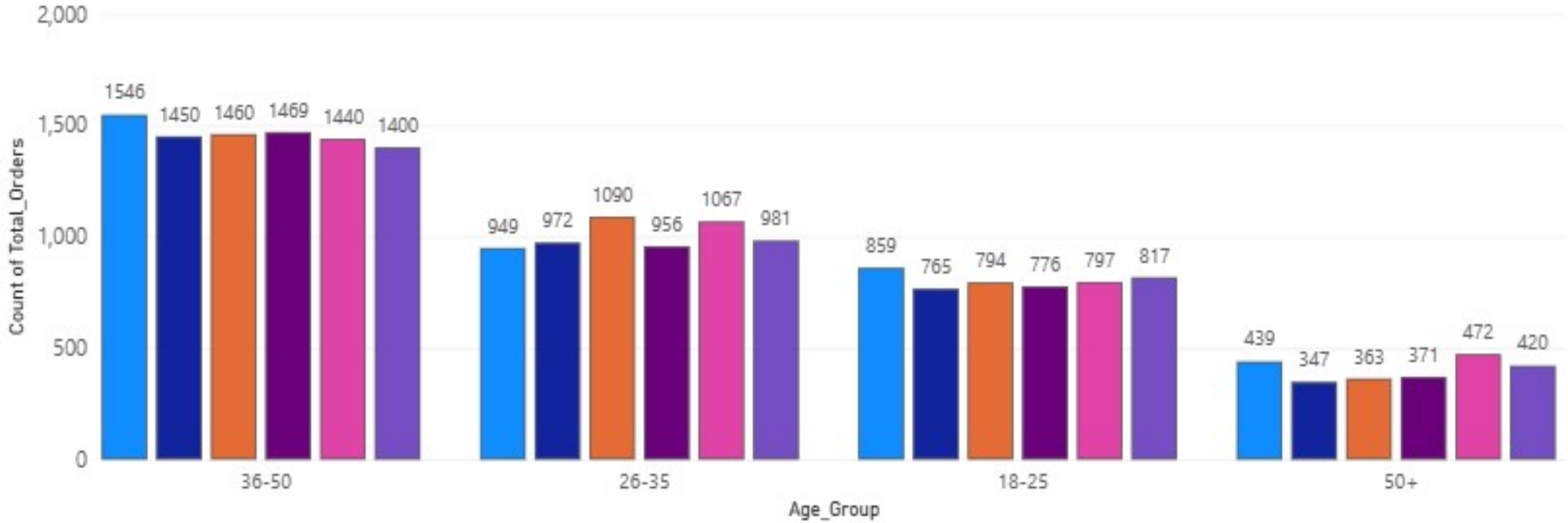
**Retention and spend:** A 70% repeat customer rate against a flat ₹559.95 average order value suggests the business retains customers well, but retained customers aren't spending meaningfully more per order than new ones — loyalty isn't yet translating into higher basket size, which could be a specific area to target (e.g., loyalty-tier discounts that push AOV up, not just repeat visits).

Count of Total_Orders
by Payment_Mode




Count of Total_Orders
by Age_Group and Preferred_Cuisine

**Age group drives order volume more than cuisine:** The 36–50 age group places noticeably more orders (~1,400–1,550 per cuisine) than any other segment, followed by 26–35 (~950–1,090), then 18–25, then 50+ (which trails far behind at 350–470). Cuisine preference within each age group is fairly evenly distributed — no single cuisine dominates any age band. age group is a real driver of order *volume*, but not of cuisine choice




**Payment mode is evenly split:** Card, Cash, UPI, and Wallet each account for almost exactly 25% of orders — no single payment method dominates. This means payment-method-specific promotions or partnerships (e.g., a UPI cashback push) wouldn't be building on an existing skew; the customer base is genuinely payment-agnostic.

# POWER BI-DASHBOARD 5

Restaurant Capacity Utilization

0.51%

City

- [ ] Urban
- [ ] Semi-Urban
- [ ] Metropolitan

Cuisine_Type

- [ ] Chinese
- [ ] Fast Food
- [ ] Indian
- [ ] Italian

| Avg_Preparation_Time_Minutes | Low_Capacity_Flag | Restaurant_ID | Avg_Rating | Restaurant Capacity U |
|---|---|---|---|---|
| 40 | Yes | R068 | 4.65 | |
| 40 | Yes | R305 | 4.60 | |
| 40 | Yes | R409 | 4.40 | |
| 40 | Yes | R496 | 4.80 | |
| 40 | Yes | R131 | 4.80 | |
| 40 | Yes | R303 | 4.50 | |
| 40 | Yes | R533 | 4.65 | |
| 40 | Yes | R061 | 4.90 | |
| 40 | Yes | R189 | 4.10 | |
| 40 | Yes | R494 | 4.80 | |
| 40 | Yes | R398 | 4.30 | |
| 40 | Yes | R407 | 4.65 | |
| 40 | Yes | R312 | 4.70 | |
| 40 | Yes | R002 | 4.10 | |
| 40 | Yes | R468 | 4.10 | |
| **Total** | | | | |

Key influencers | Top segments

What influences Restaurant Capacity Utilization to Increase ?

When...

....the average of Restaurant Capacity Utilization increases by

Avg_Rating is 4.35 or less — 58.00%

City is Semi-Urban — 46.92%

- **Capacity Flag Validated:** The "Low Capacity Flag" reliably splits the data. "Yes" restaurants are above 34.5% utilization, while "No" restaurants are below 27.8%, with zero overlap across all 550 locations.
- **Ratings Don't Matter:** Average ratings hold steady around 4.62 regardless of utilization. High-performing and underutilized restaurants have nearly identical scores, ruling out quality as the cause of the imbalance.
- **Demand-Allocation Issue:** Imbalances stem from platform demand distribution rather than restaurant performance. Over-stretched sites need delivery support or new locations, while underutilized sites need marketing pushes.
- **Data-Quality Caveat:** 31% of restaurants show utilization over 100%, which is physically impossible. This suggests the capacity metric is defined on a different scale than actual order volumes in the source data.

# POWER BI-DASHBOARD 6

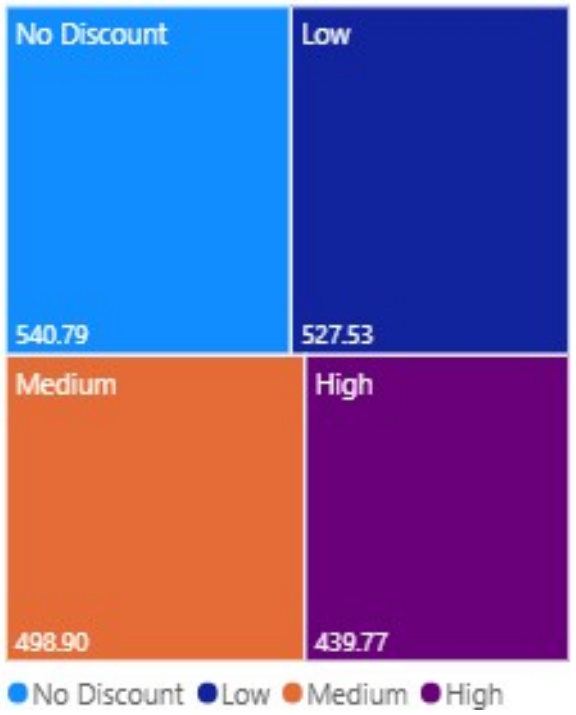
| Average Net Revenue per Order | Cancellation Rate | Total Orders | Sum of Discount_Applied | Cancelled Returned Rate |
|---|---|---|---|---|
| 520.63 | 4.95% | 100K | 2M | 7.0% |

**Higher discount levels are associated with lower Net Revenue per order, indicating that heavier discounting can erode revenue retained per order.**

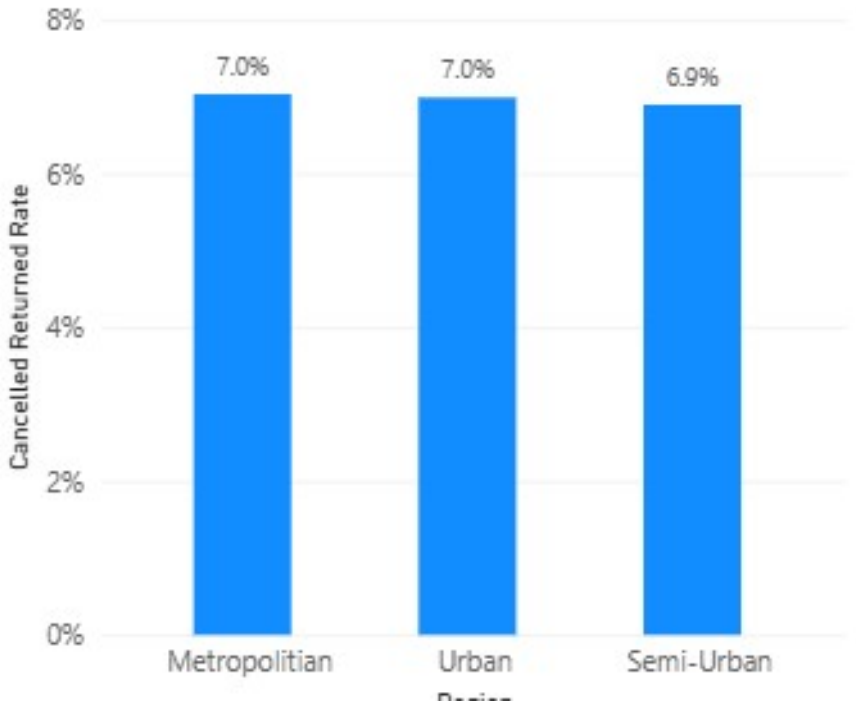
**Cancelled/returned orders do not appear to be concentrated in a particular region. The rates are broadly consistent across Metropolitan, Urban and Semi-Urban regions.**

**The vast majority of orders (93.03%) are successfully delivered. Cancellations account for 4.95% of orders, while returns represent a smaller proportion. This indicates that most orders are completed successfully, although cancellations remain an area worth monitoring.**

## Average Net Revenue per Order
by Discount Band




## Cancelled Returned Rate
by Region




## Total Orders
by Order_Status



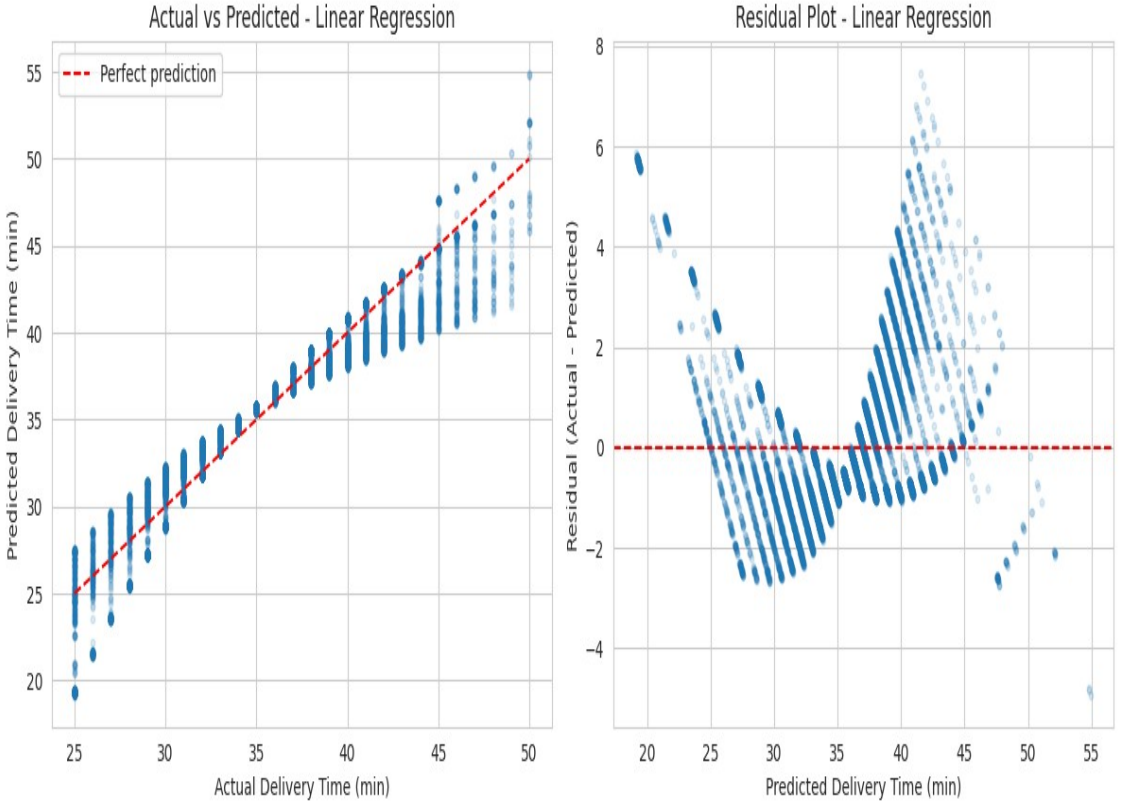
# Hypotheses & Statistical Insights

All 20 business questions were tested at α = 0.05 (ANOVA, t-test, chi-square, Pearson). Descriptive KPIs show what is happening; the inferential tests confirm which patterns are statistically real.

| Obj. | What Was Tested | Statistical Result | Decision | Business Insight |
|---|---|---|---|---|
| Obj. 1 | Region → Order Value / Revenue (ANOVA) | F=1.05, p=0.349 | Fail to Reject | Regions are statistically alike on order value |
| Obj. 1 | Festival Flag → Order Volume (t-test) | p<0.001, ~20% uplift | Reject H0 | Reliable, plannable seasonal demand lever |
| Obj. 2 | Customer Type → Avg. Spend (ANOVA) | F=199.80, p<0.001 | Reject H0 | Loyalty lifts spend ~5% (₹342.88 vs ₹326.66) |
| Obj. 2 | Age Group → Preferred Cuisine (Chi-sq.) | $\chi^2$=46.71, p<0.001 | Reject H0 | Cuisine preference is age-specific |
| Obj. 3 | Efficiency Score → Rating (Pearson) | r=0.376, p<0.001 | Reject H0 | Genuine, moderate rating driver |
| Obj. 3 | Region → Delay Rate (Chi-square) | $\chi^2$=608.21, p<0.001 | Reject H0 | Semi-Urban delay risk confirmed |
| Obj. 4 | Prep Time → Rating (Pearson) | r=−0.306, p<0.001 | Reject H0 | Actionable, fixable relationship |
| Obj. 4 | Restaurant Capacity, Metro vs. Semi-Urban | 47.42 vs. 1.27 orders/day | Descriptive | Sharp, actionable regional gap |
| Obj. 5 | Discount Depth → Net Revenue (Pearson) | r=−0.174, p<0.001 | Reject H0 | Discount ceiling recommended |
| Platform | Delivery Time & Order Value (descriptive) | 35.51 min, 89.8% on-time; ₹559.95 AOV | Descriptive | Fast, consistent service on healthy economics |

# Machine Learning Models' Insights

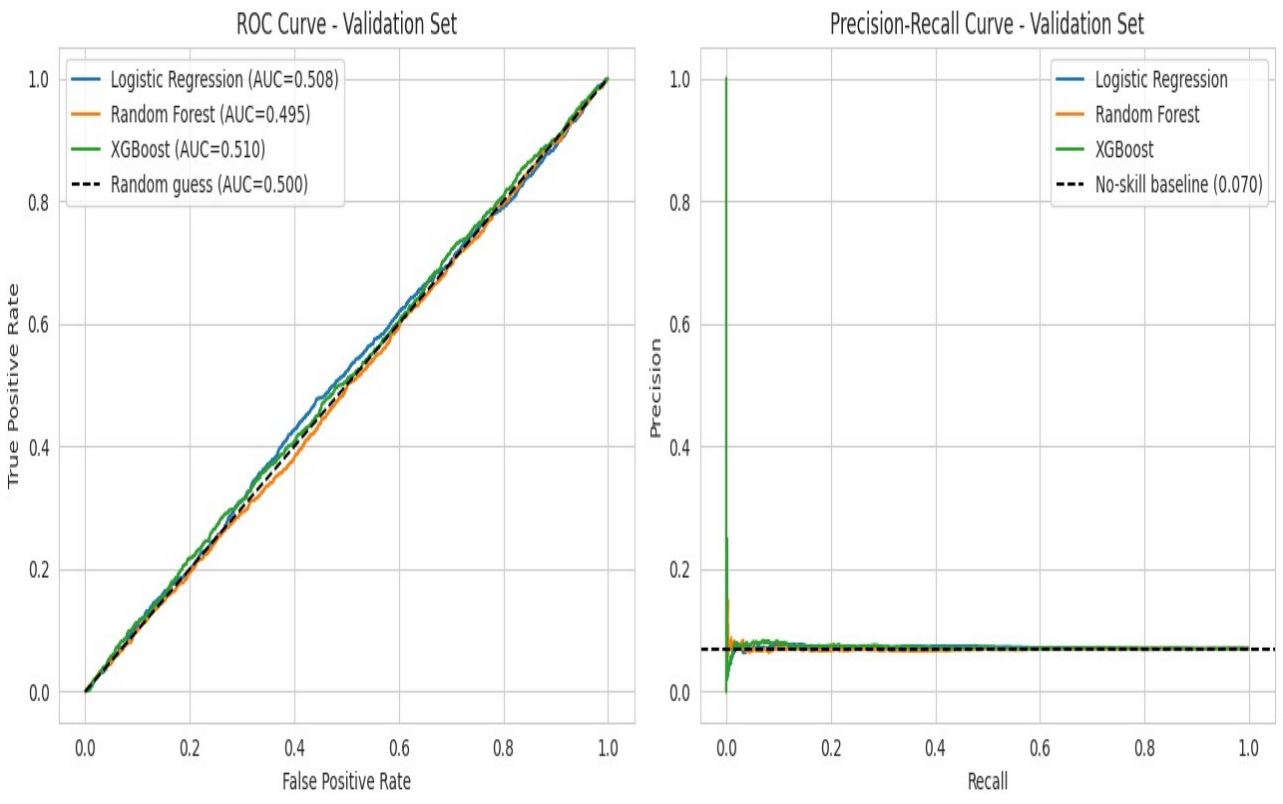
## Linear Regression — Predicting Delivery Time




**$R^2$ = 0.913, RMSE ≈ 1.45 min —** predictions track tightly along the ideal diagonal.

Delivery time is already highly explainable from FoodTech's own operational features — ready to support proactive ETA communication and staffing.

## Classification — Order-Failure Risk (Is_Failed_Order)




**AUC ≈ 0.50 across Logistic Regression, Random Forest & XGBoost —** no better than chance.

An honest, precise finding: order/customer/restaurant/partner fields alone can't separate failed from successful orders — a data-collection gap, not a modelling one.

*Delivery time is predictable enough to act on today; order-failure prediction becomes viable once GPS/traffic, kitchen-queue and payment-authorization signals are captured — until then, the Sec. 6.3/6.5 operational levers (Semi-Urban support, discount ceiling) are the fastest path to fewer failed orders.*

# Summary

**Bottom line: all three regions are evenly matched in size — the real opportunity is loyalty depth and Semi-Urban capacity, not expansion into new territory.**

### Objective 1 — Regional Demand

All three regions contribute a near-equal ~33,300 orders and ~₹1.73–1.74 crore each, but Metropolitan's repeat-order rate is 2.4–8.4× higher — proof that loyalty depth, not raw scale, is where the next unit of growth will come from.

### Objective 2 — Customer Behavior

Loyal customers spend ~10% more than Returning (₹342.88 vs ₹312.53), and cuisine preference varies significantly by age group ($p<0.001$) — a ready-made segmentation for targeted marketing.

### Objective 3 — Logistics Efficiency

Delivery Efficiency Score genuinely drives rating ($r=0.376$); delay rates differ sharply by region — pinpointing Semi-Urban as the clearest logistics investment priority.

### Objective 4 — Restaurant Performance

Prep time moderately drives rating ($r=-0.306$); capacity is concentrated in Metropolitan (47.42/day) vs. Semi-Urban (1.27) — the single biggest capacity gap on the platform.

### Objective 5 — Order Economics

Net Revenue per order falls from ₹540.79 (no discount) to ~₹430 (highest band); 93.03% of orders succeed, with cancellation risk spread evenly across regions — a policy fix, not a regional one.

# Business Implementation / Suggestion

**1.**

## Prioritize Semi-Urban capacity investment

Onboard restaurant partners and strengthen delivery-partner training/infrastructure to close its capacity and efficiency gap.

**2.**

## Introduce a discount-depth ceiling

Net Revenue per order falls measurably as discount depth increases cap it or pair it with retention goals.

**3.**

## Design loyalty-tier offers around basket size

The 70% repeat-customer rate hasn't lifted average order value target this with tiered, spend-based rewards.

**4.**

## Plan staffing around the festival calendar

The ~20% seasonal demand uplift is consistent enough across regions to plan proactive staffing in advance.

# Limitation and Future Scope

## ⚠️ Limitations

- **External cost & context drivers are unavailable —** fuel price, traffic, weather and competitor activity aren't captured, limiting the delivery-time and order-failure models.
- **Order-failure risk isn't yet reliably predictable —** with only order/customer/restaurant/partner attributes available, the classification models couldn't separate failed from successful orders (a data-availability limitation, not a modelling one).

## → Future Scope

- **Integrate real-time GPS, traffic & fuel-cost data —** to strengthen the delivery-time model and enable a future delay early-warning system.
- **Add live kitchen-queue length & payment-authorization signals —** the data-collection gap identified in the ML layer, needed for a genuinely predictive failure-risk model.

# Conclusion

- All 20 business questions across 5 objectives were answered with hypothesis-tested evidence, cross-checked across SQL, EDA, Power BI and ML.
- Demand is evenly balanced across regions — the real differentiator is loyalty depth and restaurant/delivery capacity, especially in Semi-Urban.
- Discounting and festival seasonality are the platform's two most controllable economic levers.
- Delivery time is highly predictable ($R^2 = 0.913$); order-failure prediction is a data-collection opportunity, not a limitation.
- Together, the evidence supports one platform-wide optimization strategy plus one targeted regional investment: Semi-Urban.

# THANK YOU

*Capstone Project — Analyzing Food Delivery Demand Across Regions*

Victor · Data Analytics with Generative AI — NIIT, Anna Nagar